

A new model for root growth in soil with macropores

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Abstract

Background and aims The use of standard dynamic root architecture models to simulate root growth in soil containing macropores failed to reproduce experimentally observed root growth patterns. We thus developed a new, more mechanistic model approach for the simulation of root growth in structured soil.

Methods In our alternative modelling approach, we distinguish between, firstly, the driving force for root growth, which is determined by the orientation of the previous root segment and the influence of gravitropism and, secondly, soil mechanical resistance to root growth. The latter is expressed by its inverse, soil mechanical conductance, and treated similarly to hydraulic conductivity in Darcy's law. At the presence of macropores, soil mechanical conductance is anisotropic, which leads to a

difference between the direction of the driving force and the direction of the root tip movement.

Results The model was tested using data from the literature, at pot scale, at macropore scale, and in a series of simulations where sensitivity to gravity and macropore orientation was evaluated.

Conclusions Qualitative and quantitative comparisons between simulated and experimentally observed root systems showed good agreement, suggesting that the drawn analogy between soil water flow and root growth is a useful one.

Keywords Macropores · Root architecture model · Root growth direction · R-SWMS

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Introduction

Due to high bulk densities in the subsoil, roots preferentially grow in the topsoil layer, where soil penetration resistance is low (Ehlers et al. 1983; Gregory 2008). There is, however, evidence that a significant amount of plant available nutrients as well as water supplies are stored in the subsoil. Especially during drought periods or when the top soil layer is nutrient depleted these subsoil resources play an important role in plant nutrition and can help to reduce the amount of irrigation water and fertilizer needed (Gaiser et al. 2013; Kautz et al. 2013a; Kirkegaard et al. 2007).

The extent to which plants take up nutrients and water from the subsoil essentially depends on the fraction of roots that are able to penetrate this hard soil layer

(Kuhlmann and Baumgärtel 1991). A possibility for roots to gain access to deeper, highly dense soil horizons is to use large sized macropores (diameters >2 mm) as preferential pathways (Ehlers et al. 1983; Kautz et al. 2013b; McKenzie et al. 2009; Stewart et al. 1999; Stirzaker et al. 1996). The probability of roots to grow in macropores depends on the abundance of pores in the soil (Hatano et al. 1988) and on the penetration resistance of the bulk soil (Hirth et al. 2005). While some studies (Stewart et al. 1999; Stirzaker et al. 1996) observed that significantly more roots encountered macropores than what would be expected if root growth was purely random, others (Dexter 1986; McKenzie et al. 2009) assume that roots locate macropores only by chance. In greater depths, where the effect of overburden pressure causes high soil penetration resistances, Gao et al. (2016a) state that root growth in macropores should be considered the norm rather than the exception. (Kautz et al. 2013b) observed that roots use macropores to overcome hard soil layers, but then again re-enter the bulk soil. This is in line with the results by Hirth et al. (2005) who found roots to grow more frequently in macropores when the bulk density is higher. Dexter and Hewitt (1978), Stirzaker et al. (1996) and Hirth et al. (2005) observed that roots tend to grow over a longer distance in macropores that are aligned more vertically. Hatano et al. (1988), Stirzaker et al. (1996) and Valentine et al. (2012) have shown that root elongation in macropores is higher than in the surrounding bulk soil.

Roots do not only use macropores as preferential pathways, but also take up nutrients from the pore walls, which were observed to be rich in nutrients (Athmann et al. 2014). Due to the generally low water content inside macropores when soil is dry (Laloy et al. 2010), root water uptake from the pore walls is vital (White and Kirkegaard 2010). Knowledge about the root – macropore – soil contact is thus essential. Athmann et al. (2013) have shown that the way roots connect to the pore wall depends on the plant genotype. White and Kirkegaard (2010) and Kautz and Köpke (2009) found most roots to grow straight through the pore and connect to the pore wall by the help of root hairs respectively lateral branches if they do not have direct contact. Athmann et al. (2013) observed that barley roots spiral down in large coils inside the pore wall. Field studies have shown that 85% of the roots of a barley and oilseed rape crop, which were found in macropores established contact to the pore wall (Athmann et al. 2013).

These plant scale observations converge with our current understanding how environmental stimuli influence root growth. Toyota and Gilroy (2013) physiologically analyzed the mechanisms of gravitropic and mechanical signaling in roots. Shkolnik et al. (2016) state the importance of hydrosensing, where roots grow away from low water potential towards higher water potential. Bao et al. (2014) observed that the formation of lateral roots depends on the availability of water in the vicinity of the root.

The influence of macropores on root growth as well as on root water and nutrient uptake from the subsoil is hard to measure directly. Simulation models that describe root development in structured soils and water and nutrient fluxes in the root zone are therefore useful tools to interpret measurements that provide indirect information about uptake processes, e.g. soil water contents, plant nutrient contents and water and nutrient isotopic profiles in the soil and in the plant. Until now, only few models exist, which include the responses of roots to macropores (Vereecken et al. 2016). Gaiser et al. (2013) modeled the effect of macropores on root development at the plot scale and Jakobsen and Dexter (1988) investigated the influence of macropores on root growth and water uptake in a water balance model. In these model simulations, the amount of roots that grow into macropores was prescribed or parameterized. But how this parameterization changes with changing soil properties (e.g. matric bulk density, amount and orientation of macropores) and root growth parameters (e.g. root growth responses to soil penetration resistance, gravity) cannot be predicted by these models but is required model input. Such predictions require explicit simulation of root growth and development at both the single root and the root system scale.

In recent years, several different simulation models for the description of growing root systems have been developed. While the early models merely focused on the representation of the root system architecture, the later models are more complex and also include the influence of the surrounding soil. Most of these later models (Clausnitzer and Hopmans 1994; Pagès et al. 2004) calculate the rate and direction of root growth as the vector sum of various root segment length and direction-affecting components. Root growth models frequently use the concept of tropisms to represent the influence of plant physiological properties on the direction of root growth. The gradient of the environmental stimulus that triggers a certain tropism defines the

direction in which the root tip will grow. Most root growth models include the influence of gravitropism (Clausnitzer and Hopmans 1994; Leitner et al. 2010; Pagès et al. 2004) and some also take into account chemotropism and hydrotropism (Leitner et al. 2010; Tsutsumi et al. 2003). The effect of soil heterogeneities on the direction of root growth is typically implemented similarly to the concept of tropisms: The gradient of soil mechanical resistance defines the direction of root growth, i.e. roots grow in the direction in which the soil resistance decreases most rapidly (Clausnitzer and Hopmans 1994; Pagès et al. 2004). The influence of soil mechanical resistance on the root growth direction is controlled by a sensitivity factor. However, using this ‘tropism approach’, we were not able to simulate root growth along the macropore wall or to simulate the way that roots appear to ‘find’ macropores in deeper soil layers and grow into them. Whilst the approach is logical and attractive in simulating root growth in bulk soil, it needs to be modified to enable the simulation of root growth in, along, and out of macropores. This is primarily due to large gradients in strength, geometry, and matric potential that change rapidly adjacent to macropore walls. Therefore, a more mechanistic description of root growth to determine the root growth direction seems necessary.

This study presents a new method for computing root growth in soils with macropores. It distinguishes between the driving forces for root growth and anisotropy of soil strength, which is similar to the description of water flow in a soil with anisotropic hydraulic conductivity. The new ‘anisotropy approach’ is illustrated by the simulation of an experimental study by Stirzaker et al. (1996) on root growth in artificial macropores. Experimental and simulation results using both the tropism and anisotropy approach to model changes in root growth direction are compared visually and quantitatively. The potential of the anisotropy approach to simulate the effects of different macropore inclination angles, bulk soil penetration resistances, and gravitropism on root growth in structured soil are demonstrated by comparing simulation results with experimental data from Hirth et al. (2005). To evaluate its performance on root growth in a multi-layered soil domain containing macropores, we carried out a simulation study that was inspired by an experimental study by Dexter (1986).

Material and methods

Model description

In our new approach, we draw an analogy between the movement of a root tip and water flow in porous media (Bear 2013). The root tip is pushed by a root inherent driving force into the soil. The direction of this driving force depends on the direction of the previous root segment and on a gravitational component that is directed downwards. This driving force is counteracted by soil mechanical forces and friction. If the friction force depends on the direction of the movement, i.e. when the friction or soil resistance is anisotropic, the movement of the root tip will deviate from the direction of the driving force.

In analogy with soil water flow, the direction of the root tip movement corresponds to the water flux vector, while the driving force represents the gradient of the water potential. Soil mechanical forces can be seen as the viscous friction forces that counteract water flow and thus the driving force. In the Darcy equation, the effect of these viscous forces on energy dissipation is represented by the hydraulic conductance tensor, which may show anisotropy. Analogous to the Darcy flow equation, we express the movement of the root tip by the following equation:

$$\mathbf{d} = \mathbf{k} * \mathbf{F}, \quad (1)$$

where $\mathbf{d}$ is root tip movement vector, $\mathbf{k}$ is the soil mechanical conductance tensor that represents the ease with which the root can penetrate the soil and $\mathbf{F}$ is the driving force that influences the root growth direction.

The soil mechanical conductance tensor $\mathbf{k}$ is a symmetric, second rank tensor with nine entry values defining conductances in the three principal directions:

$$\mathbf{k} = \begin{pmatrix} k_{xx} & k_{xy} & k_{xz} \\ k_{yx} & k_{yy} & k_{yz} \\ k_{zx} & k_{zy} & k_{zz} \end{pmatrix}. \quad (2)$$

For an isotropic soil domain, where soil penetration resistance, or conductance as its inverse, is uniform in each direction, the conductance tensor $\mathbf{k}$ can be reduced to a diagonal matrix in which all diagonal entry values are identical. The direction of movement of the root tip then merely depends on the driving force. The simplest example of soil heterogeneity is a stratified soil domain

where each layer has a different conductance. In accordance with soil hydraulic conductivity, the soil mechanical conductance in direction of the soil layering equals the arithmetic mean whereas the conductance perpendicular to the soil layering is equal to the harmonic mean of the individual soil layer conductances. If the direction of the soil layers, i.e. the axis of anisotropy, coincides with one of the axes of the Cartesian coordinate system, the conductance tensor is a diagonal matrix with three different entry values. If the layering or the axes of anisotropy are not aligned with the Cartesian coordinate system, the conductance tensor $\mathbf{k}$ is fully occupied with nine entry values. In an anisotropic medium, the root tip movement deviates from the direction of the driving force and is oriented towards the axis of anisotropy in which the conductance is largest and resistance is smallest. Anisotropy in the soil domain arises if big differences in soil density occur on a small spatial scale and can thus be caused by macropores, soil aggregation or differently compacted soil layers. Furthermore, local differences in soil water content e.g. due to root water uptake, also lead to anisotropy and affect the direction of root growth. Unlike the approaches by Clausnitzer and Hopmans (1994) and Pagès et al. (2004), no sensitivity factor is needed to weigh the influence of penetration resistance on the root growth direction.

The driving force $\mathbf{F}$ could comprise several factors. We chose here the direction vector of the previous root segment and gravitropism. The direction of the previous root segment is expressed by the azimuth angle α and the polar angle β . To account for small scale variations in the soil matrix and to represent a random behavior of the root tip, random deflection angles γ and δ are added to α and β (Fig. 1, Eq. (3)).

$$\mathbf{F} = \begin{pmatrix} dx_{(\alpha, \beta, \gamma, \delta)} \\ dy_{(\alpha, \beta, \gamma, \delta)} \\ dz_{(\alpha, \beta, \gamma, \delta)} \end{pmatrix} + sg \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}; \quad (3)$$

While the first term on the right hand side of Eq. (3) represents the previous growth direction vector of $\mathbf{F}$, the second term expresses the gravitropism component with sg as gravitropism sensitivity factor.

While the deflection of the azimuth angle γ is a uniformly distributed random angle between $[0 \ 2\pi]$, the deflection of the polar angle δ is a normally distributed random angle with mean zero and standard

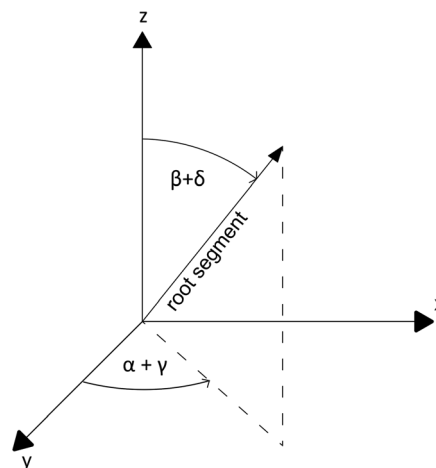


Fig. 1 Direction of the root segment expressed by the azimuth angle α with random deflection γ and the polar angle β with random deflection δ

deviation σ_{dl} , which is calculated following the approach by Leitner et al. (2010). The standard deviation σ_{dl} is derived from the user defined unit standard deviation σ of a root segment of 1 cm length and the maximum root segment length dl , which is reached when soil penetration resistance equals zero. In probability theory, standard deviation decreases by the square root of the number of trials. If $1/dl$ (segments per cm) is regarded as the number of trials, the standard deviation σ_{dl} ($^{\circ} \text{ cm}^{1/2}$) can be given as

$$\sigma_{dl} = \sqrt{dl} * \sigma. \quad (4)$$

In this way, the deflection from the original root tip location does not depend on the spatial resolution of the root growth model. By using the maximum root segment length as normalization factor for the standard deviation of the random deflection angle, we create a dependency between σ_{dl} and soil penetration resistance. In this way, experimental observations of higher root tortuosity in more compact soil (Tracy 2013) are taken into account.

Mathematical formulation of the anisotropy approach to model changes in root growth direction

We implemented our new model approach in the three dimensional numerical R-SWMS model (Javaux et al. 2008). This model couples the root growth model by Clausnitzer and Hopmans (1994) with a model that simulates water flow in the soil domain and in the root system (Doussan et al. 1998; Richards 1931). For the numerical solution of the water flow equation, the soil

domain is discretized in a regular cubic grid of nodes. Initial soil hydraulic properties and soil bulk density are user defined input values and given explicitly for each node of the grid. The system of root branches consists of straight root segments. At each root growth time step, a new root segment emerges from the tip of a growing branch and moves the root apex to a new position. Length and orientation of the newly developed root segment are influenced by a soil mechanical conductance tensor and a root inherent driving force (Eq. (1)).

Soil penetration resistance in the bulk soil is calculated for each grid node as a function of soil bulk density, soil water potential and effective saturation using the pedotransfer function developed by Whalley et al. (2007):

$$\log_{10}R = 0.35 * \log_{10}(|\psi| * S_e) + 0.93 * \rho_b + 1.26, \quad (5)$$

where R is the soil penetration resistance (kPa), ψ the water potential (kPa), S_e the effective saturation (–) and ρ_b the soil bulk density (g cm^{-3}). This pedotransfer function is based on the analysis of 12 different soils with varying bulk density and organic carbon, sand, silt and clay contents from 3 different soil depths (5, 15, 25 cm) and can thus be assumed to be valid for a wide range of soils. It has, however, the limitation of not taking into account the effect of overburden pressure due to increasing soil depth. For simulations of soil domains with greater depths, it is recommended to use a pedotransfer function that includes the effect of overburden pressure such as described in Gao et al. (2016b). Soil mechanical conductance k (kPa^{-1}) is then determined as the inverse of soil penetration resistance R :

$$k = \frac{1}{R}. \quad (6)$$

Soil penetration resistances respectively soil mechanical conductances are specified at each node of a grid cell. We assume that these nodal soil mechanical conductances are direction independent or isotropic properties. The eight nodal conductances of the grid cell in which the root tip is located are used to determine an average or grid cell conductance tensor (Eq. (2)), which is assumed to be constant within the grid cell. Homogeneous nodal conductances lead to an isotropic soil mechanical conductance.

In a regular cubic soil grid, macropores are designed in a stepwise structure by arranging grid cells on top or next to each other (Fig. 2). For whichever inclination angle of the macropore, the principal axes of anisotropy then either coincide with or stand at an angle of 45° to one of the three axes of the Cartesian coordinate system. If macropores are the cause of soil anisotropy, it is thus sufficient to only consider four possible orientations of the axes of anisotropy. Rotating the Cartesian coordinate system by 45° around each one of its main axes gives us three local coordinate systems of anisotropy (Fig. 3). For each root tip, we then calculate four different conductance tensors and choose the one for which the contrast between the main axes of anisotropy is largest.

In the simplest case where the main axes of anisotropy coincide with the axes of the Cartesian coordinate system, the conductance tensor is calculated as follows: In all three directions of the Cartesian coordinate system, the grid cell is virtually cut into two halves, which are regarded as two separate soil layers with different conductances. The average soil conductance of each half space of one grid cell is calculated as the arithmetic mean of the conductance values of the four corner nodes located within this half (Fig. 4, Eq. (7)):

$$k_{h_{x1}} = \frac{k_1 + k_2 + k_3 + k_4}{4}. \quad (7)$$

Each axis of the Cartesian coordinate system is aligned perpendicular to two half spaces of a grid cell. In line with hydraulic conductivity, the average

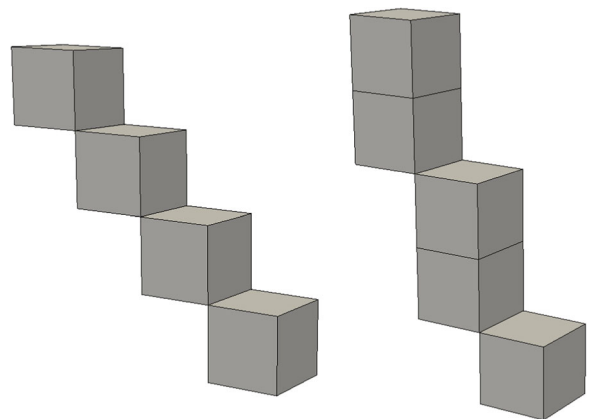


Fig. 2 Stepwise structure of a macropore, which is inclined by 45° respectively by 60° from the horizontal

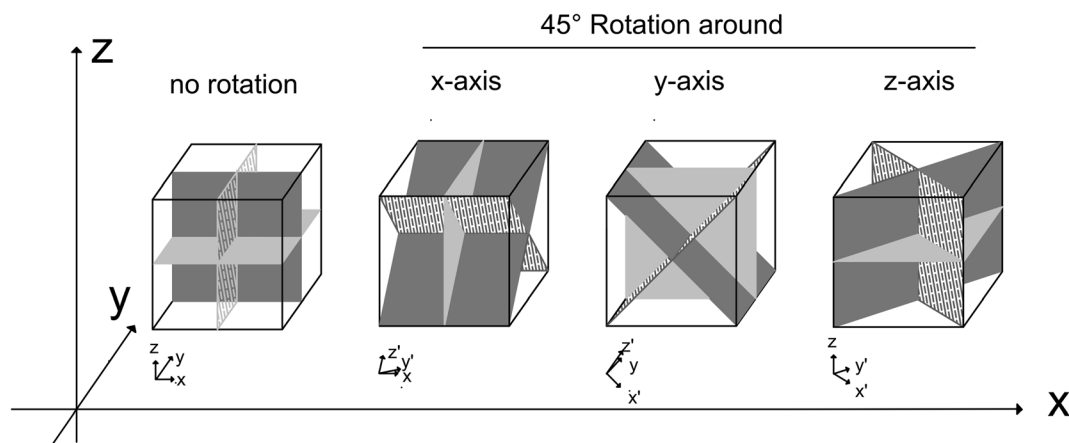


Fig. 3 Four local coordinate systems are sufficient to describe the possible orientations of the main axes of anisotropy in a regular cubic soil grid. The planes perpendicular to the local coordinate

axes are used to divide one cubic soil element in two half-spaces that are needed to compute local average conductances (e.g. Figure 5)

conductance for each direction is calculated as the harmonic mean of the conductances of two opposing half-spaces of a grid cell (Fig. 5, Eq. (8)).

$$k_{xx} = \frac{2}{\frac{1}{kh_{x1}} + \frac{1}{kh_{x2}}} \quad (8)$$

In case that the main axes of anisotropy do not coincide with axes of the Cartesian coordinate system, we calculate the average soil conductance for each grid cell half perpendicular to the axes of the rotated coordinate system. The average soil conductance of the half space on either side of a rotated plane is the arithmetic mean of six weighted

conductances: While the conductances of the two corner nodes, which lie within one half are given the weight 1, the conductances of the four corner nodes lying on the separating plane between two halves have the weight 0.5. The average conductance for each direction is then once again calculated as the harmonic mean of the conductances of two opposing grid cell halves. Figure 6 and Eqs. (9), (10) and (11) give an example for the calculation of the conductance in y' – direction of the coordinate system that was rotated around the x -axis.

$$k'_{y1} = \frac{k_1 + k_2 + \frac{k_3}{2} + \frac{k_4}{2} + \frac{k_5}{2} + \frac{k_6}{2}}{4}, \quad (9)$$

$$k'_{y2} = \frac{k_7 + k_8 + \frac{k_3}{2} + \frac{k_4}{2} + \frac{k_5}{2} + \frac{k_6}{2}}{4}, \quad (10)$$

$$k'_{yy} = \frac{2}{\frac{1}{k'_{y1}} + \frac{1}{k'_{y2}}}, \quad (11)$$

where k'_{y1} and k'_{y2} are the average conductances of the rotated halves of a grid cell in y' direction, k_1 to k_8 are the conductance values of the corner nodes of the grid cell and k'_{yy} is the conductance vector of the local coordinate system, which was rotated around the x -axis in y' direction.

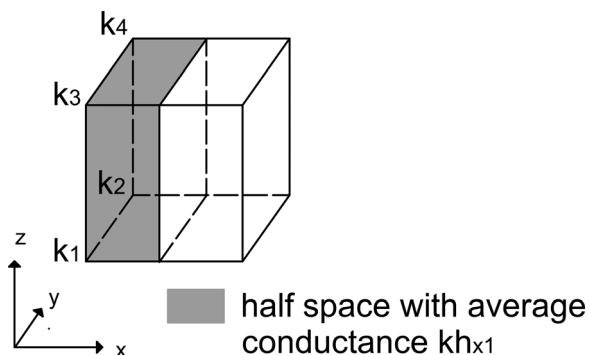


Fig. 4 Average conductance of one half space of a grid element perpendicular to the x -axis

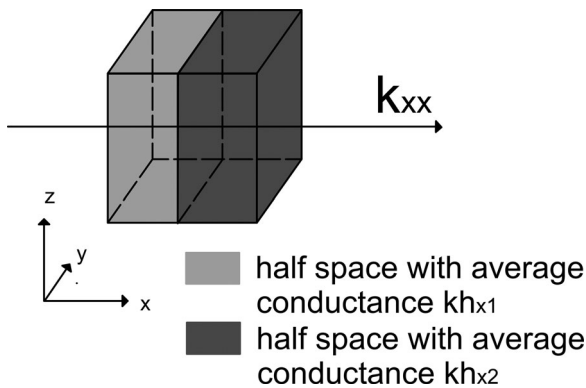


Fig. 5 Conductance perpendicular to the conductances of the two half spaces

The three conductance tensors in their local coordinate systems are then mapped back onto the Cartesian coordinate system by the help of a rotation matrix (Eq. (12) following the approach by Lust (2001):

$$\mathbf{k} = \mathbf{M}_{\text{rot}} * \mathbf{k}' * \mathbf{M}_{\text{rot}}', \quad (12)$$

where $\mathbf{k}$ is the conductance tensor in the Cartesian coordinate system, $\mathbf{k}'$ is the conductance tensor in the local coordinate system, $\mathbf{M}_{\text{rot}}$ is the rotation matrix and $\mathbf{M}_{\text{rot}}'$ is the conjugated rotation matrix.

The length of a newly developed root segment is calculated as the product of root elongation rate and a user defined root growth time step. The root elongation rate E_r (cm d^{-1}) is assumed to be a function of soil strength. Bengough et al. (2011) observed that the soil penetration resistance sufficient to stop root elongation

completely (R_{max} , kPa) is a function of matric potential (ψ , kPa) and can be calculated as

$$R_{\text{max}} = 4000 + 2.33 * |\psi|, \quad (13)$$

In the bulk soil, R_{max} corresponds approximately to the soil penetration resistance at the permanent wilting point ($\psi = -1500$ kPa). The root elongation rate is assumed to decrease linearly between zero and maximum soil penetration resistance. The actual root elongation rate can thus be calculated as a fraction of the maximum root elongation rate as

$$E_r = E_{\text{max}} * \left(1 - \frac{R_{\text{eff}}}{R_{\text{max}}}\right), \quad (14)$$

where E_r (cm d^{-1}) is the actual root elongation rate, E_{max} (cm d^{-1}) is the maximum root elongation rate and R_{eff} is the effective soil penetration resistance in the direction of the root tip growth. R_{eff} is by definition the inverse of an effective conductance in the direction of root growth k_{eff} :

$$R_{\text{eff}} = \frac{1}{k_{\text{eff}}}; \quad (15)$$

k_{eff} is a function of both the average conductance of the grid cell and the root inherent growth direction and is calculated as:

$$k_{\text{eff}} = |\mathbf{k} * \hat{\mathbf{F}}|, \quad (16)$$

where $\mathbf{F}$ is the unit length vector of the driving force. In this way, E_r is reduced stronger if the root grows perpendicular to a hard soil layer than if it grows along a hard soil layer. This approach corresponds to observations by Kolb et al. (2012) who found that radial constrictions applied to roots did not significantly reduce root elongation rates, while axial constrictions did have a significant impact.

Model assumptions for root growth in macropores

In the case of a root growing within a grid cell at the interface between bulk soil and macropore, soil domain and macropore are regarded as two soil regions with different soil mechanical conductances that influence the direction in which the root will grow. The intensity with which a root is forced to grow towards the direction of higher conductance depends on both the conductances in the bulk soil and in the macropore. While the conductance in the bulk soil is

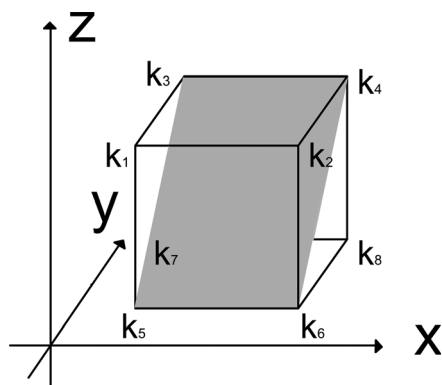


Fig. 6 Separating plane between two halves perpendicular to the y' – direction of the local coordinate system which was rotated around the x – axis

calculated as the inverse of soil penetration resistance, the conductance in the macropore (k_{macro}) is unknown. To identify plausible values of k_{macro} , we analyzed the anisotropy of one single grid cell with four bulk soil and four macropore nodes. We define the degree of anisotropy (DA) according to Dal Ferro et al. (2014) as

$$DA = 1 - \frac{k_{\text{perp}}}{k_{\text{long}}}, \quad (17)$$

where k_{perp} and k_{long} are the conductances perpendicular and along the macropore – bulk soil grid cell half spaces (Fig. 7). A DA of 0 signifies perfect isotropy, while a DA of 1 represents maximum anisotropy. Figure 8 shows the influence of different parametrizations of k_{macro} on anisotropy for typical minimum and maximum values of soil penetration resistance. Assuming that the conductance in the macropore is much higher than in the bulk soil, anisotropy shall be well above 0. If the degree of anisotropy approaches 1, however, the influence of different soil conductances is no longer perceptible. We therefore assume macropore conductance values of $1e4 \leq k_{\text{macro}} \leq 2e5 \text{ kPa}^{-1}$ as most plausible. The conductance in the macropore can be regarded as a sensitivity factor, which influences the probability of a root to continue growing within the macropore or to re-enter the bulk soil.

Model setup

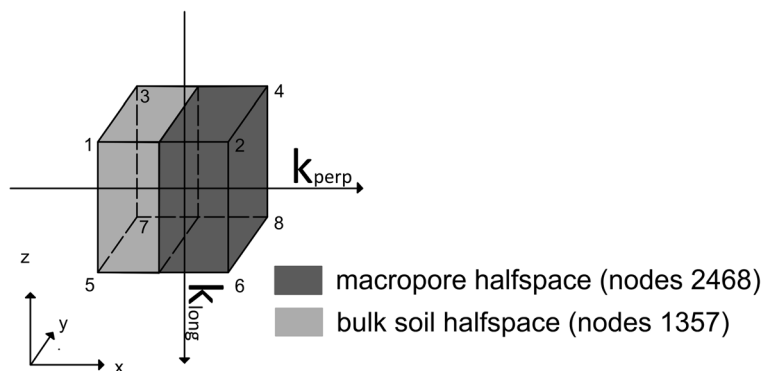
Using experimental model setups from literature, we built three simulation scenarios to assess the performance of our new model approach.

Scenario 1: visual comparison of simulation results with observed root growth patterns in structure soil

For our first simulation scenario, we used an experimental study by Stirzaker et al. (1996) on root growth of barley (*Hordeum vulgare* cv. Yagan) in soil containing macropores at the plant root scale. The setup of the simulation model was designed according to the descriptions by Stirzaker et al. (1996). Undescribed model parameters were either taken from literature or approximated. The soil domain was a rectangular cuboid with a surface area of $8.7 \times 8.7 \text{ cm}^2$ and a depth of 20.1 cm, which we discretized to cubic grid elements of 0.1 cm side length. The bulk density of the boundary grid cells of the sides and the base of the soil domain were set to a virtual density of 4 g cm^{-3} in order to simulate the impenetrable pot walls. Eight vertical macropores with a diameter each of 0.4 cm were arranged symmetrically around the center of the soil domain on a circle with a radius of 2.5 cm (Fig. 9). We used the soil properties of a sandy loam (Table 1), which was packed to a bulk density of 1.77 g cm^{-3} . We did not consider soil water flow and assumed hydrostatic equilibrium in the soil domain. The simulation runtime was set to 25 days.

The simulated root system consisted of seven axes from which one emerged at day zero, three at day one and three at day three. The initial potential root elongation rate for barley was derived from literature (Materchera et al. 1991) and set to 1.2 cm d^{-1} . Watt et al. (2006) observed the growth rates of roots to decrease with time and branch roots to grow more slowly than their parent axes. We thus reduced the elongation rate for 8 day old first order roots to 0.8 cm d^{-1} . Root images by Stirzaker et al. (1996) show that roots grew in a low angle from the horizontal over the whole width of the pot before they

Fig. 7 The degree of anisotropy is one minus the ratio between the conductance perpendicular to (k_{perp}) and along (k_{long}) the plane that separates the macropore from the bulk soil half space



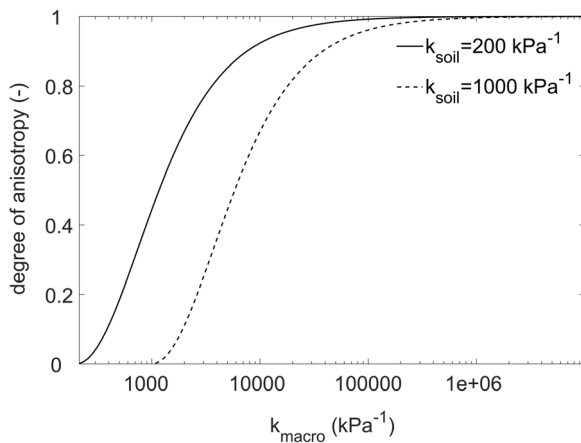


Fig. 8 Influence of k_{macro} on the degree of anisotropy for typical minimum and maximum values of k_{soil}

turn downwards (Fig. 11 (a)). In order to reproduce these root growth patterns, sensitivity to gravitropism was set to the extremely low value of 0.005 for 1st order roots. Rose (1983) observed roots of higher branching to be less gravitropic. Sensitivity to gravitropism for 2nd and 3rd order roots was thus reduced to 0.001. The root tortuosity as displayed in Fig. 11 (a) could be best reproduced with unit standard deviations of the random angle of 45° . The initial growth angle for axial roots was set to 0° ; the branching angle (relative to the parent roots) to 90° . Branch spacing was estimated and set to a value of 0.6 days for 1st order roots and 0.4 days for 2nd order roots.

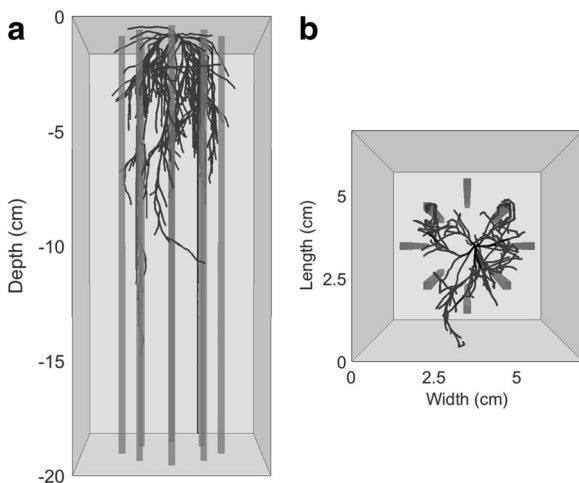


Fig. 9 Side (a) and top (b) view of the soil domain with a 25 – day old barley root, scenario 1; the bulk soil is displayed in light grey, while the macropores are presented in dark grey and the root in black

Root water uptake was not considered in the simulations. Soil water potentials were set so that simulated soil penetration resistances matched experimentally measured ones. We chose a whole pot matrix potential of -100 kPa, which resulted in soil penetration resistances of 2500 kPa and corresponded approximately to the experimentally observed values by Stirzaker et al. (1996), which lay between 2000 and 4000 kPa. The complete parameter set is presented in Table 2, while the values for the different parametrizations are presented in Table 3.

Scenario 2: quantitative comparison of simulation and experimental results on single root growth in inclined macropores

For our second simulation scenario, we used an experimental study by Hirth et al. (2005) on the ability of seedling roots of rye grass (*Lolium perenne* L.) to penetrate the soil from artificial macropores under varying soil bulk densities and macropore inclination angles. The setup of the simulation model was designed according to the descriptions by Hirth et al. (2005). Undescribed model parameters were either taken from literature or approximated. The rectangular-shaped soil domain had a surface area of 3×1 cm² and a depth of 3 cm, which we discretized to cubic grid cells of 0.1 cm side length. One single macropore with an angle of 40° respectively 90° was inserted into the soil domain. Macropore and interface had a horizontal cross section area of 0.09 cm². The seed (starting point of the root tip) was placed at the edge of the macropore (Fig. 10). We used the soil properties of a silty loam with the texture indicated in Table 1. The soil was packed to uniform bulk densities of 1.25, 1.38 and 1.50 g cm⁻³. In accordance with the experimental setup, macropore wall compaction was not considered. Hirth et al. (2005) kept the matric potential in the soil cores at a constant value of -5 kPa

Table 1 Texture of soils used in the simulation scenarios 1, 2 and 3

Simulation scenario	FAO soil classification	Sand (%)	Silt (%)	Clay (%)
Scenario 1	Sandy loam	74	12	14
Scenario 2 and 3	Silty loam	55.8	26.6	12.3

Table 2 Model parametrizations for simulation scenarios 1, 2 and 3

Geometry of the soil domain				Root growth parameters									Simulation runtime	
L (cm)	W (cm)	D (cm)	ρ_b (g cm ⁻³)	ψ (kPa)	$E_{\max}$ (cm d ⁻¹)	nbas (-)	basang (°)	brang (°)	brspac (d ⁻¹)	σ (°)	sg (-)	k_{macro} (kPa ⁻¹)	(d)	
Scenario1	8.7	8.7	20.1	1.77	-100	1.2 (0.8)	6	0	90	0.6 (0.4)	45	0.005 (0.001)	2.00E + 05	25
Scenario2	3	1	3	Table 3	-5	0.68	-	-	-	-	Table 3	Table 3	Table 3	7
Scenario3	3	3	3	1.25 resp. 3	-15	0.68	-	-	-	-	45	0.05	1.00E + 05	10

Values in parentheses indicate parametrizations for 2nd and 3rd order roots, L Length, W Width, D Depth, ρ_b bulk density, ψ soil matric potential, $E_{\max}$ maximum elongation rate, n_{bas} number of basal roots, $basang$ basal root angle, $brang$ branching angle, $brspac$ branch spacing, σ unit standard deviation of the random angle, sg sensitivity to gravitropism, k_{macro} conductance in the macropore

by connecting them to 0.5 m hanging columns of water. We therefore assumed that root water uptake does not significantly affect the surrounding soil and performed simulations without root water uptake. We did not consider soil water flow and assumed hydrostatic equilibrium in the soil domain. The simulation runtime was set to 7 days.

The simulation of only one single root without laterals reduced the required input parameters for root growth to potential root elongation rate, sensitivity to gravitropism, unit standard deviation of the random deflection angle and conductance in the macropore. A potential root elongation rate of 0.49 cm d⁻¹, which we assumed to stay constant over time was best suited to reproduce the actual root lengths measured by Hirth et al. (2005). This value is within the range of the standard error of the mean of the potential root elongation rate for seedlings of annual ryegrass (*L.rigidum*) given by Materechera et al. (1991). The remaining root growth parameters were not experimentally determined and thus unknown. To evaluate the influence of different root growth parametrizations, we performed simulations with different combinations of these parameters (see Table 3 for chosen parameter values). Altogether, we carried out 576 different simulations, which were the factorial combinations of three bulk densities, two macropore angles and a control soil domain without macropore, four sensitivities to gravitropism (sg), four unit standard deviations of the random deflection angle (σ) and four conductances in the macropore (k_{macro}). The complete parameter set is presented in Table 2.

To obtain representative simulation results of the stochastic process, which is generated by the random deflection angle, we performed 100 replicates of each simulation using different random seed numbers. Experimental results reported by Hirth et al. (2005) represent the average of 24 replicates, but no information of standard deviations was provided.

Scenario 3: virtual simulation experiment on root growth in a compacted subsoil layer that contains macropores

For our third simulation scenario, we carried out a simulation experiment on single root growth in a two-layered soil domain where the compacted subsoil contained macropores. The simulation experiment represents the case of root growth in soil with a plough layer. The soil domain was a rectangular cuboid of 3 cm

Table 3 Values for different parametrizations of scenario 2

Inclination angle of the MP (°)	Soil bulk density, ρ_b (g cm^{-3})	Root growth parameters		
		Sensitivity to gravitropism, sg (–)	Unit stdev of the random angle, σ (°)	Conductance in the MP, k_{macro} (kPa^{-1})
No macropore	1.25	0.005	5	$2.00\text{E} + 4$
40	1.38	0.05	45	$3.00\text{E} + 4$
90	1.50	0.1	90	$5.00\text{E} + 4$
		0.2	180	$8.00\text{E} + 4$

side length in each direction, which we discretized to cubic grid cells of 0.1 cm side length. We implemented two different soil layers into this soil cube: a topsoil layer with a bulk density of 1.25 g cm^{-3} in the upper 1.5 cm of the cube and an impenetrable subsoil layer with a bulk density of 3 g cm^{-3} in the lower 1.5 cm of the cube. Additionally, we inserted eight macropores into the compacted subsoil layer. They were aligned symmetrically in a square with a distance of 0.9 cm to the borders of the soil cube (Fig. 15). Each macropore cross section was made up of nine grid nodes with macropore properties. We used root growth parameters from the previous example for rye grass (*Lolium perenne* L.) with a sensitivity to gravitropism of 0.05 and a unit standard deviation of the random angle of 45° . Root water uptake was not included. We did not consider soil water flow and assumed hydrostatic equilibrium in the soil domain. The simulation runtime was set to

10 days. We performed only one simulation. The complete parameter set is presented in Table 2.

Results

Scenario 1: visual comparison of simulation results with observed root growth patterns in structure soil

We used both the tropism and anisotropy approach in order to simulate the experimental observations by Stirzaker et al. (1996). The quality of the simulation results was evaluated visually by comparing 2D-images of the simulated and the experimental root systems and quantitatively by comparing total root lengths and root length density profiles (RLD profiles).

Both approaches led to simulation results where the roots predominantly did use macropores as preferential growth pathways. The root growth behavior within macropores, however, was different: Using the tropism approach, the roots only slowly grew downwards while spiraling horizontally over the whole cross section of the macropore; using the anisotropy approach, the roots grew straight downwards along the pore wall, which better captures experimental observations (Fig. 11). The simulation results in Fig. 11 are displayed in a layout so as to resemble the original figure from Stirzaker et al. (1996), Fig. 6c. We found a total root length of 750 cm for the simulated root system, which corresponds well to the experimentally observed one of 720 cm.

We determined the RLD profile from the original 2D image from Stirzaker et al. (1996), Fig. 6c with the help of the image analysis tool Root System Analyzer (Leitner et al. 2014) and compared it with

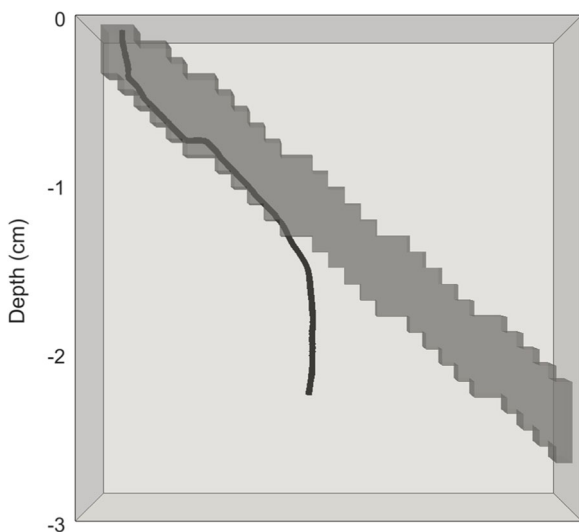
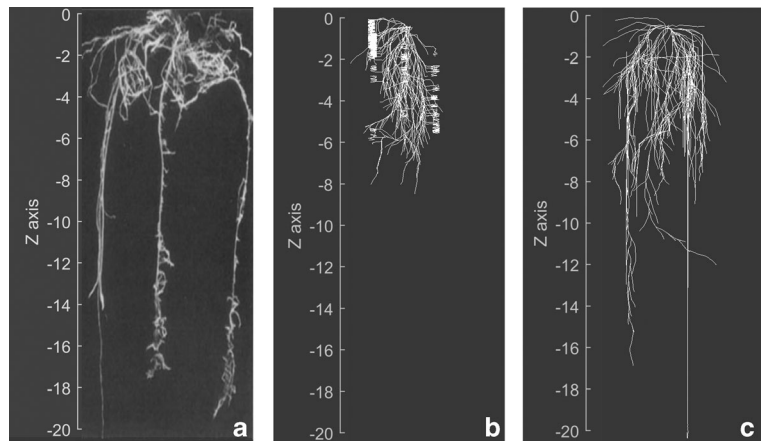


Fig. 10 Soil domain, scenario 2; the bulk soil is displayed in light grey, while the macropores are presented in dark grey and the root in black

Fig. 11 Front view of barley roots growing in dense soil with macropores for 25 days: (a) Experimental results by Stirzaker et al. (1996), (b) Simulation results produced with the tropism approach, (c) Simulation results produced with the anisotropy approach



the RLD profiles of the simulated 3D root systems (Fig. 12). It must be noted that the RLD profile obtained from the 2D image from Stirzaker et al. (1996) can only be an approximation of the RLD profile of the real root system due to low image resolution and the two dimensional representation of a three dimensional root system. The RLD profile produced with the anisotropy approach was able to capture the larger root length density in the upper 5 cm of the soil domain, which then decreased sharply. The RLD profile produced with the tropism approach largely overestimated RLD in the upper soil domain, while underestimating it in the lower soil domain. The root length density within macropores (area between the dashed and the solid line) as a percentage of total RLD was similar for the experimental RLD profile (26%) and the simulated RLD profile produced with the anisotropy approach (21%).

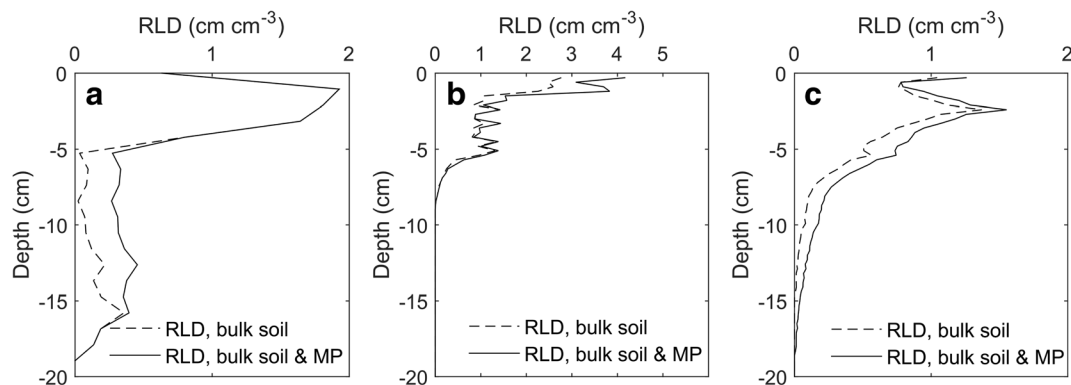


Fig. 12 Root length density profiles of barley roots growing in dense soil with macropores for 25 days: (a) RLD profile for original 2D image by Stirzaker et al. (1996), (b) RLD profile for

Scenario 2: quantitative comparison of simulation and experimental results on single root growth in inclined macropores

Simulation and experimental results by Hirth et al. (2005) were compared quantitatively. We used the characteristics total root length (cm) and root length fraction that remained within the macropore (%) as a means of comparison between experiment and simulation. The variability of the averaged results of different simulations is caused by different parameter combinations, while the variability of the individual results is the random variation between the 100 replicate simulations.

Influence of different macropore inclination angles and different bulk densities

Figure 13 shows a comparison between the simulation results obtained with a randomly chosen parametrization

simulated 3D root system produced with the tropism approach, (c) RLD profile for simulated 3D root system produced with the anisotropy approach

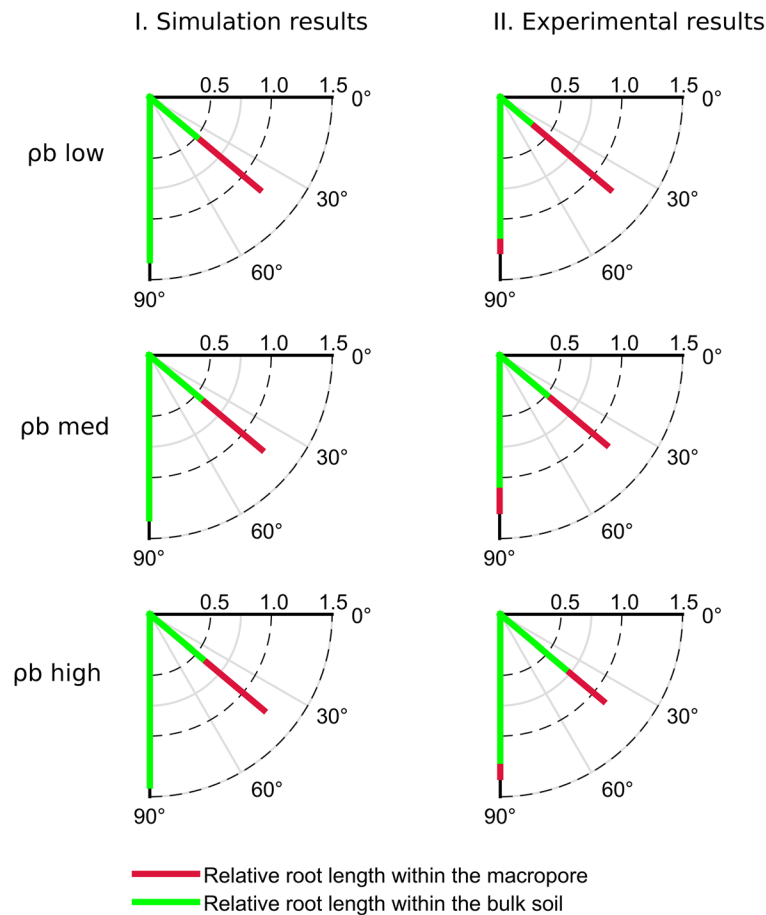


Fig. 13 Simulated and experimentally found relative root lengths within macropore and bulk soil; the first column (I) shows the simulation results obtained with a randomly chosen parametrization ($sg = 0.05$, $\sigma = 45^\circ$, $k_{macro} = 8e4 \text{ kPa}^{-1}$), while the second (II) column illustrates the experimental results by Hirth et al. (2005). The different rows show results for different levels of soil bulk density (pb low, pb med, pb high). The inclination angles of the colored lines represent the macropore inclination angles (40° , 90°);

($sg = 0.05$, $\sigma = 45^\circ$, $k_{macro} = 8e4 \text{ kPa}^{-1}$), and the experimental results by Hirth et al. (2005) for a smooth macropore wall. The simulations captured well the experimental observations of increasing root length fractions within the macropore with an increasing macropore inclination angle from the horizontal. In accordance with the experimental observations, different levels of bulk density only had an effect on the roots growing in the 40° inclined macropores. Simulations were able to reproduce the experimentally observed increase in root length fractions within the macropore for increasing levels of bulk density. Due to this increase, total root lengths did not decrease for increasing levels of bulk density. Compared to root growth in a homogeneous soil domain

the different colors indicate the different locations of the root within the soil domain (macropore, bulk soil). The length of the colored lines represents the relative root length which is the total root length normalized with the length of a root growing in a soil domain with equal bulk density, but without macropore. Each line in the first column represents the average of 100 individual simulations. Each line in the second column represents the average of 24 individual simulations

with equal bulk density, the presence of macropores increased total root lengths by 20% to 40%.

Quantitative simulation results: 90° - inclined macropore

In the case of a 90° inclined (vertical) macropore, all simulated roots remained within the macropore for all different parameter combinations. They reached average root lengths between 2.8 cm and 3.3 cm and thus grew at 83% to 98% of the potential root elongation rate. These results correspond well to the findings by Hirth et al. (2005) who measured average root length fractions in

macropores between 83% and 90% and total root lengths between 2.9 cm and 3.1 cm.

Quantitative simulation results: 40° - inclined macropore

In the case of a macropore with 40° inclination, the simulated average root length fraction within a macropore reached – depending on the parametrization - values between 18% and 60%. Simulated averaged total root lengths ranged from 2.5 cm to 3.0 cm. The root thus grew at 74% to 89% of the potential root elongation rate. Hirth et al. (2005) found – depending on the roughness of the macropore walls – root length fractions within macropores between 14% and 86% and total root lengths between 1.9 cm and 3.0 cm. Both simulated root length fractions within macropores and total root lengths were thus in acceptable agreement with experimental results.

Influence of different parametrizations on the simulation results of the 40° inclined macropore

Figure 14 gives an overview of the influence of different parametrizations (Table 3) on simulated root length fractions remaining within a 40° inclined macropore. Different parametrizations of bulk density and conductance in the macropore (k_{macro}) were pooled in the parameter ‘degree of anisotropy’ (Fig. 7, Eq. (17)).

To evaluate the influence of different degrees of anisotropy on root length fractions remaining within the macropore, we fitted linear regression lines to the simulation results of each parameter combination. As expected, increasing degrees of anisotropy led to an increase in root length fractions within a macropore for nearly all parameter combinations. The coefficients of determination show that the variability of the simulation results increased both with increasing standard deviations of the random angle and decreasing sensitivities to gravitropism. For parameter combinations including a sensitivity of gravitropism of 0.005, no regression line could be fitted due to the high variability of the simulation results. There is a trend of increasing root length fractions and decreasing rates of increase both with increasing standard deviations of the random angle and decreasing sensitivities to gravitropism, but the pattern is not consistent. For individual simulations, root length fractions within macropores of up to 100% could be

reached; the maximum value for the intercept of a regression line with a degree of anisotropy of 1, however, was only 50%.

Influence of the roughness of macropore walls

Hirth et al. (2005) performed experiments with smooth and scarified macropore wall reliefs. They found significant differences in both root length fractions within macropores and total root lengths for the two different treatments. Larger root length fractions remained within the smooth macropore (averaged over all bulk density levels, 68%) than within the scarified macropore (averaged over all bulk density levels, 38%). Consequently, total root lengths were larger for roots growing in smooth macropores (on average 2.85 cm) than for roots growing in scarified macropores (on average 2.3 cm). In the simulation model, it is not possible to directly take into account macropore wall roughness. However, the influence of wall roughness can be controlled indirectly via the conductance in the macropore. In the parametrization example from Fig. 13, an increase of k_{macro} from $2\text{e}4 \text{ kPa}^{-1}$ to $8\text{e}4 \text{ kPa}^{-1}$ led to an increase in the average root length fraction within the macropore from 33% to 44% if averaged over all bulk density levels.

Scenario 3: virtual simulation experiment on root growth in a compacted subsoil layer that contains macropores

The simulation result produced with the anisotropy approach captured well the expected root growth behavior (Fig. 15b). When reaching the compacted subsoil layer, the root grew horizontally along it keeping constant contact to the soil layer until it encountered a macropore. It then entered the pore and grew straight down along the pore wall. Using the tropism approach (Fig. 15a), the root was not able to enter the macropore, but oscillated around its opening without entering it.

Discussion

Concepts of root growth models and their parameters are difficult or even impossible to validate or derive from direct measurements. However, by comparing simulated root architectures with experimentally observed ones, different concepts can be compared with each other and more appropriate ones can be

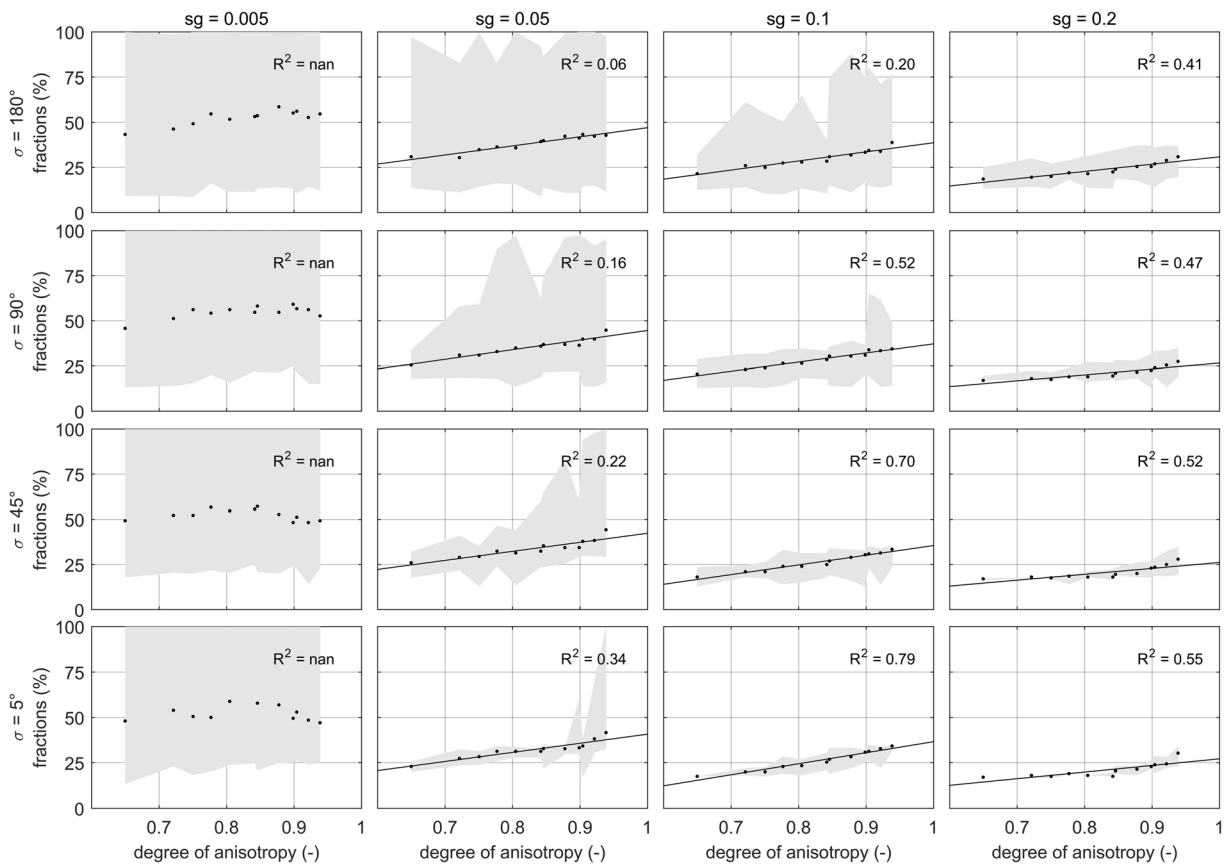


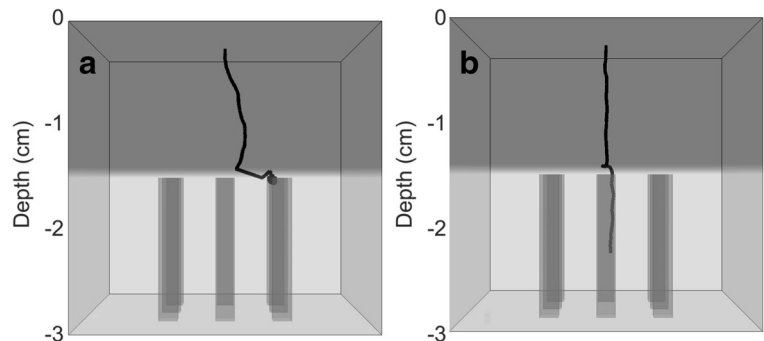
Fig. 14 Influence of different parametrizations of sensitivity to gravitropism (sg), unit standard deviation of the random angle (σ) and degree of anisotropy on the fractions of root lengths remaining within a 40° inclined macropore. Each of the 16 figures shows a different combination of sg and σ . The black dots show the

averaged root length fraction that remains within the macropore for each degree of anisotropy; the grey shaded area shows the hull of the total range of the simulation results. R^2 specifies the coefficient of determination of the linear regression line that was fitted to the total range of the simulation results

identified. In this study we demonstrated that our new anisotropy approach to simulate changes in root growth direction due to soil penetration resistance is more appropriate to describe the development of root systems in soil with macropores. Data from experiments in which parameters like the macropore

inclination angle and the matric bulk density were systematically varied could be used to constrain parameters of the root growth model. It must be noted that these experiments were not designed with the purpose of calibrating or validating a root growth model. Using a simulation model to design or plan

Fig. 15 Soil domain and root simulated with the tropism (a) and the anisotropy (b) approach; the topsoil layer is presented in dark grey, the subsoil layer in light grey; the macropores are displayed in light grey and the root in black



such experiments could be beneficial to measure variables that contain additional information and allow a better determination of model parameters. In this example, the variability of root lengths and root length fractions within macropores could have been an additional source of information since it differed strongly between simulations using different parameter values.

Hirth et al. (2005) showed that the roughness of macropore walls has a significant influence on the probability of a root to continue growing within a macropore or to re-enter the bulk soil. For the simulation of roots growing in natural macropore networks, knowledge about the macropore wall roughness is thus mandatory. Combined information on root growth in macropores and on the roughness of earthworm burrow walls or root channels such as provided by Leue and Gerke (2016) could be used to calibrate the simulation model in that way.

Macropore walls were observed to be richer in nutrients than the surrounding bulk soil (Athmann et al. 2014; Barej et al. 2014; Jiménez et al. 2003). For simulations of root growth in macropores including nutrient uptake, information about the root – macropore wall contact is thus essential.

The new model approach was developed for a simulation domain that is discretized into a regular cubic grid of nodes. If used in a model with a different description of the simulation domain, an alternative approach must be devised for the calculation of soil mechanical conductances. An example for such a model is RootBox (Leitner et al. 2010), where soil physical properties are not assigned to grid nodes, but implemented in a lattice-free way using signed distance functions.

For simulations of root growth in macropores on a scale larger than the single root scale, we rely on extensive experimental data to parametrize and validate the model. Non-invasive methods such as rhizotron studies (Kuchenbuch and Ingram 2002; Nagel et al. 2012; Tracy et al. 2010) may provide reliable information on root elongation rates and root growth parameters. Imaging methods such as X-ray computed tomography (Rab et al. 2014; Tracy et al. 2010) or magnetic resonance imaging (Gruwel 2014; Stingaciu et al. 2013) can be used to characterize the spatial distribution of both macropore networks and plant roots in 3D and additionally to visualize and quantify soil water dynamics including preferential flow (Sammartino et al. 2015).

Our new anisotropy approach to model the effect of macropores on root growth direction is part of the mechanistic 3D model for water and solute transport in the soil-root system, R-SWMS (Javaux et al. 2008). Thus, it is a contribution to a better understanding of underlying processes and feedback loops of soil – plant interactions on the root system scale.

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